

Evan Landman
Jarrett Walker &
Associates

R for Public Transit Analysis

Developing reproducible transit analysis in R in a consulting environment.

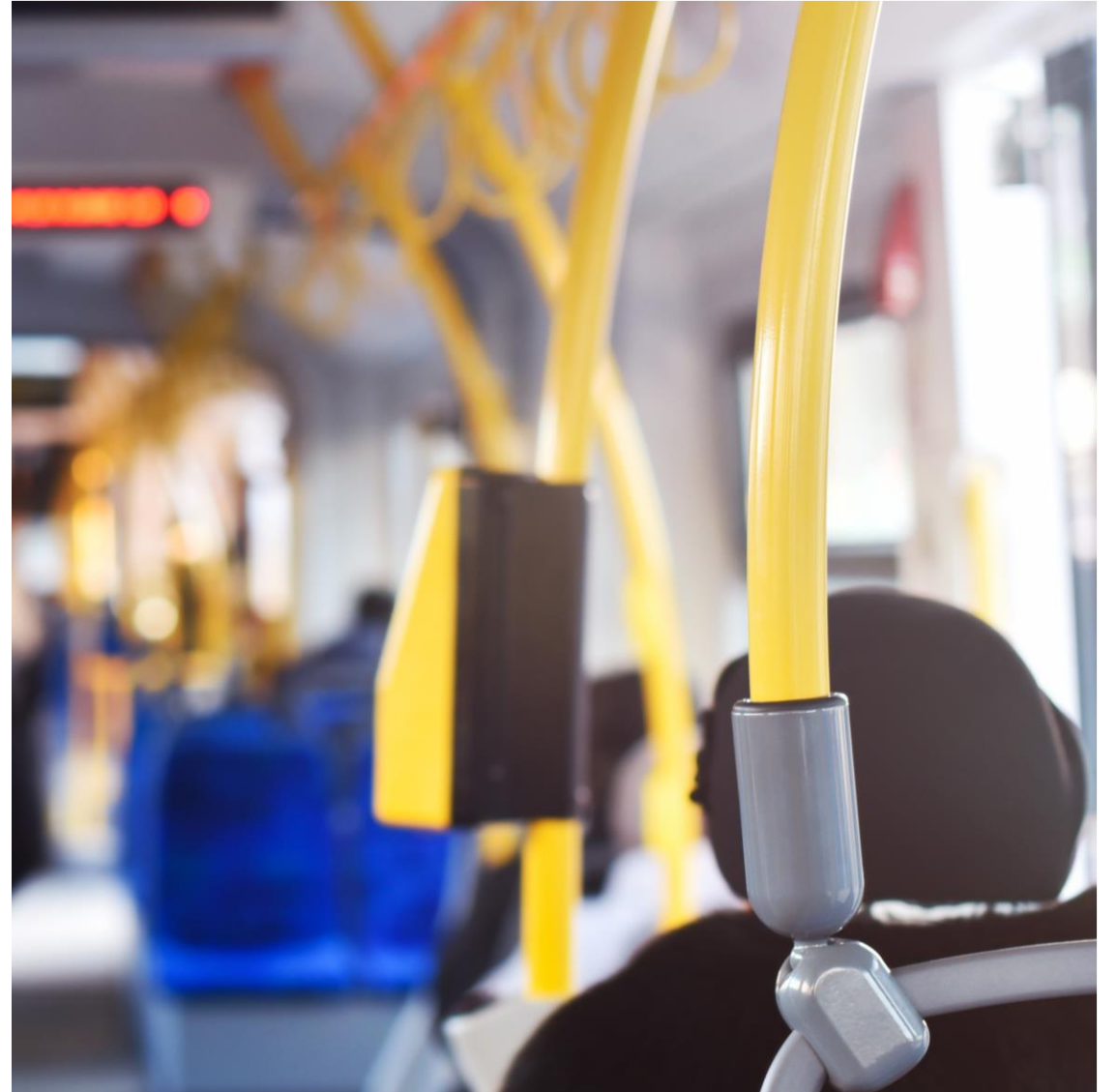


TRANSIT

That thing that you sometimes use to get to work, or meet a friend, or go to the store.

And... a complex spatial process.

And... data-rich industry with endless opportunity to do interesting things with R.



Transit Planning

Transit agencies are constantly evaluating their service.

- How can we make our service more reliable?
- How can we make our service easier to use?
- How can we attract more riders?

We support agencies looking to study the design of their network.

You've seen our work if you've ridden the bus in Portland, Vancouver, Dallas, Houston, San Jose, Cleveland, and many other cities.

Transit planning is political.

- Transit agencies constantly hear conflicting things from the public.

"Your service sucks!"

"Don't change anything!"

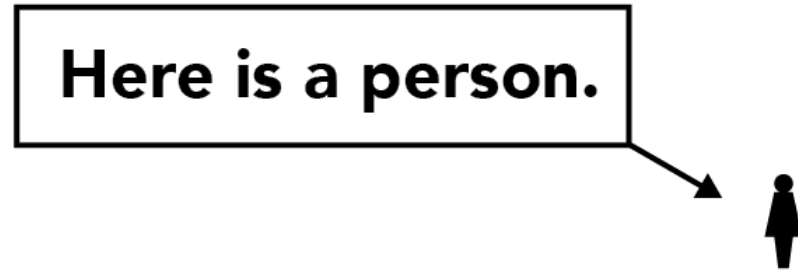
- Particularly in the US, we basically never have enough money to meet every need.



Our Job

Help transit agencies make good decisions about what to prioritize with their service.

How do we make transit useful?



JARRETT
WALKER
+ ASSOCIATES

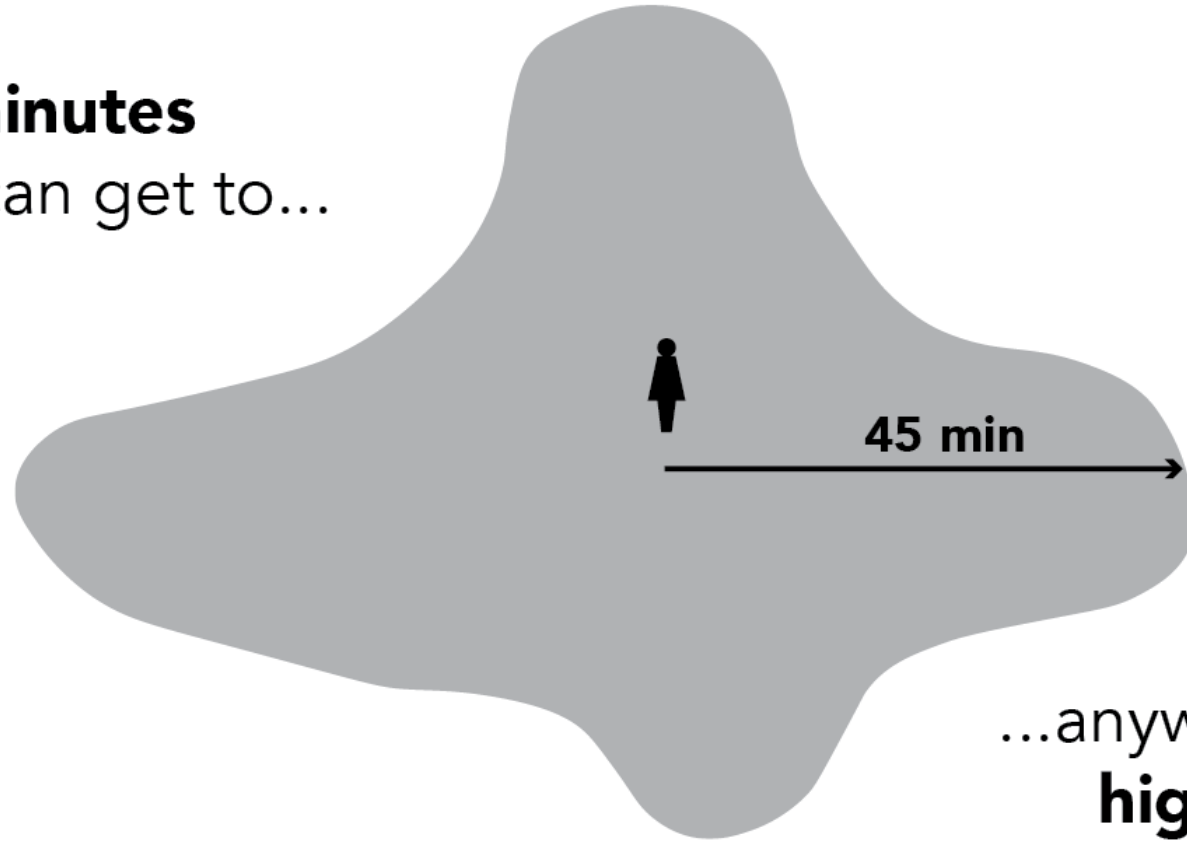
Where do you need to go?

She is in **a city full of possible destinations.**

JARRETT
WALKER
+ ASSOCIATES

Where can transit take you?

In **45 minutes**
she can get to...



...anywhere in the
highlighted area.

JARRETT
WALKER
+ ASSOCIATES

Which opportunities can you access?

Her **access to destinations** is the **number of destinations in that area.**

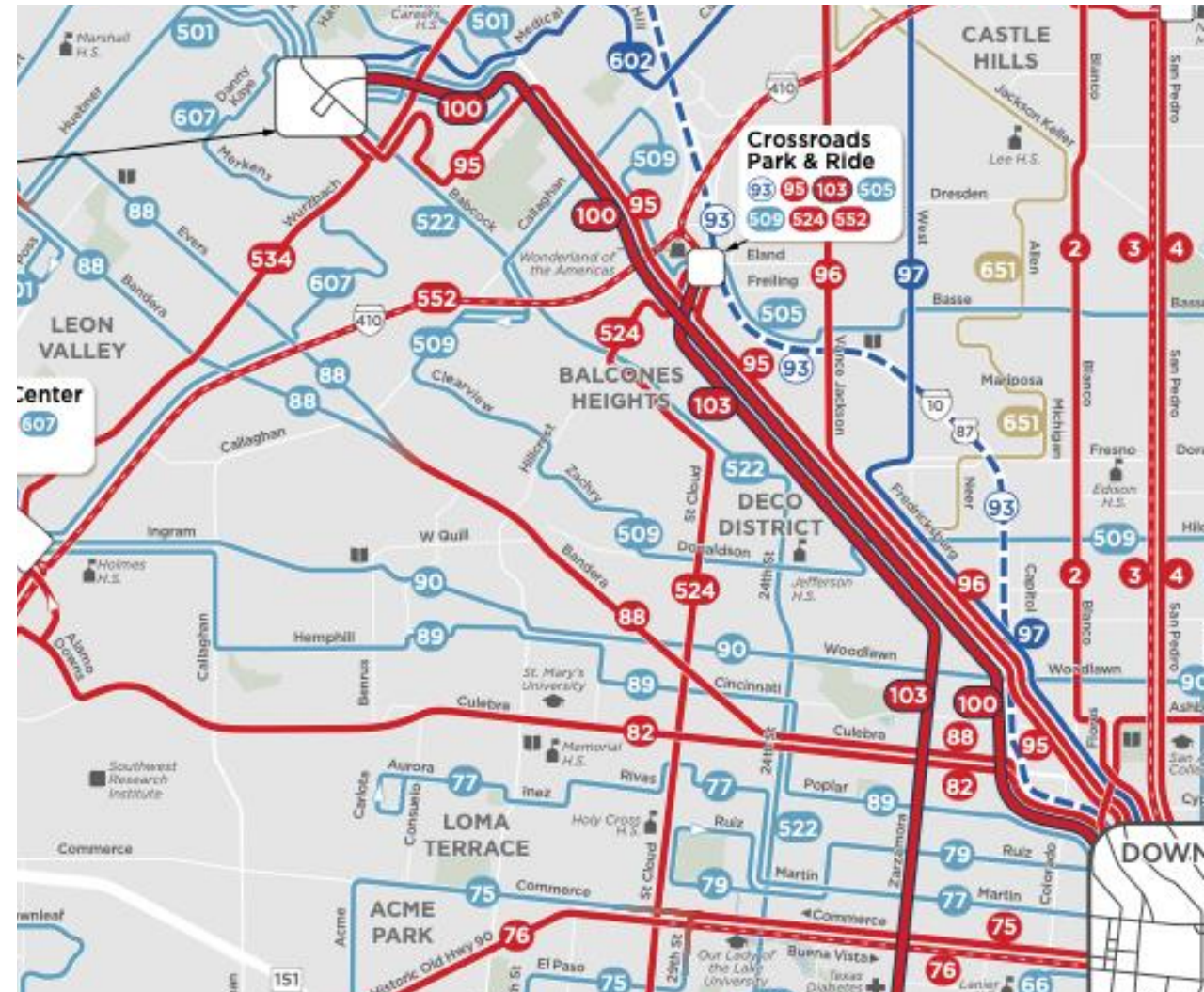
JARRETT
WALKER
+ ASSOCIATES



You can count the jobs or schools or shopping in that area to estimate her access.

These are data-informed questions.

- Where do people need to go?
- Where is it convenient or inconvenient to use transit?
- What changes could make things more useful?
- How much would it cost?
- How would people be impacted?



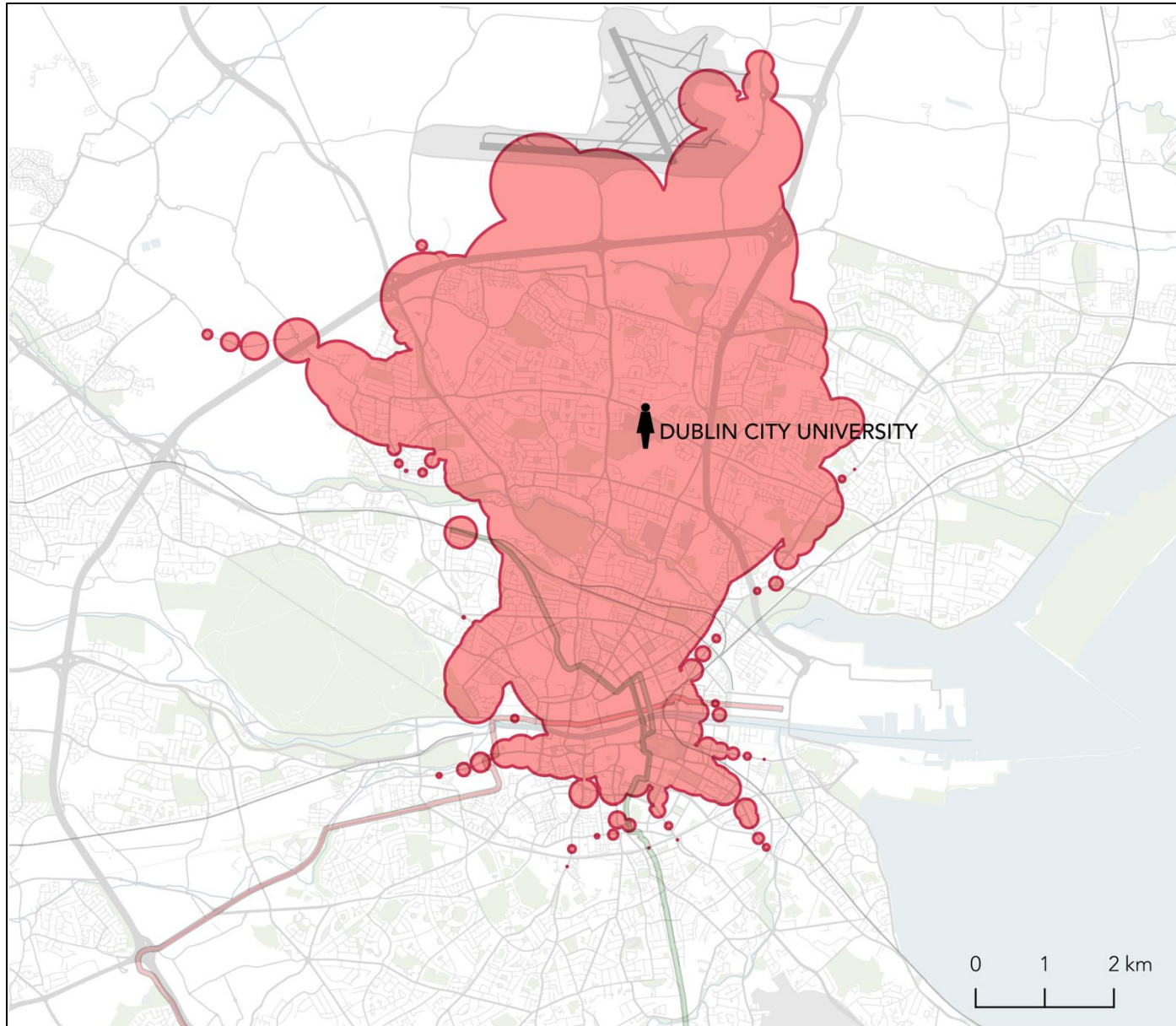
Where can I go in
45 minutes?



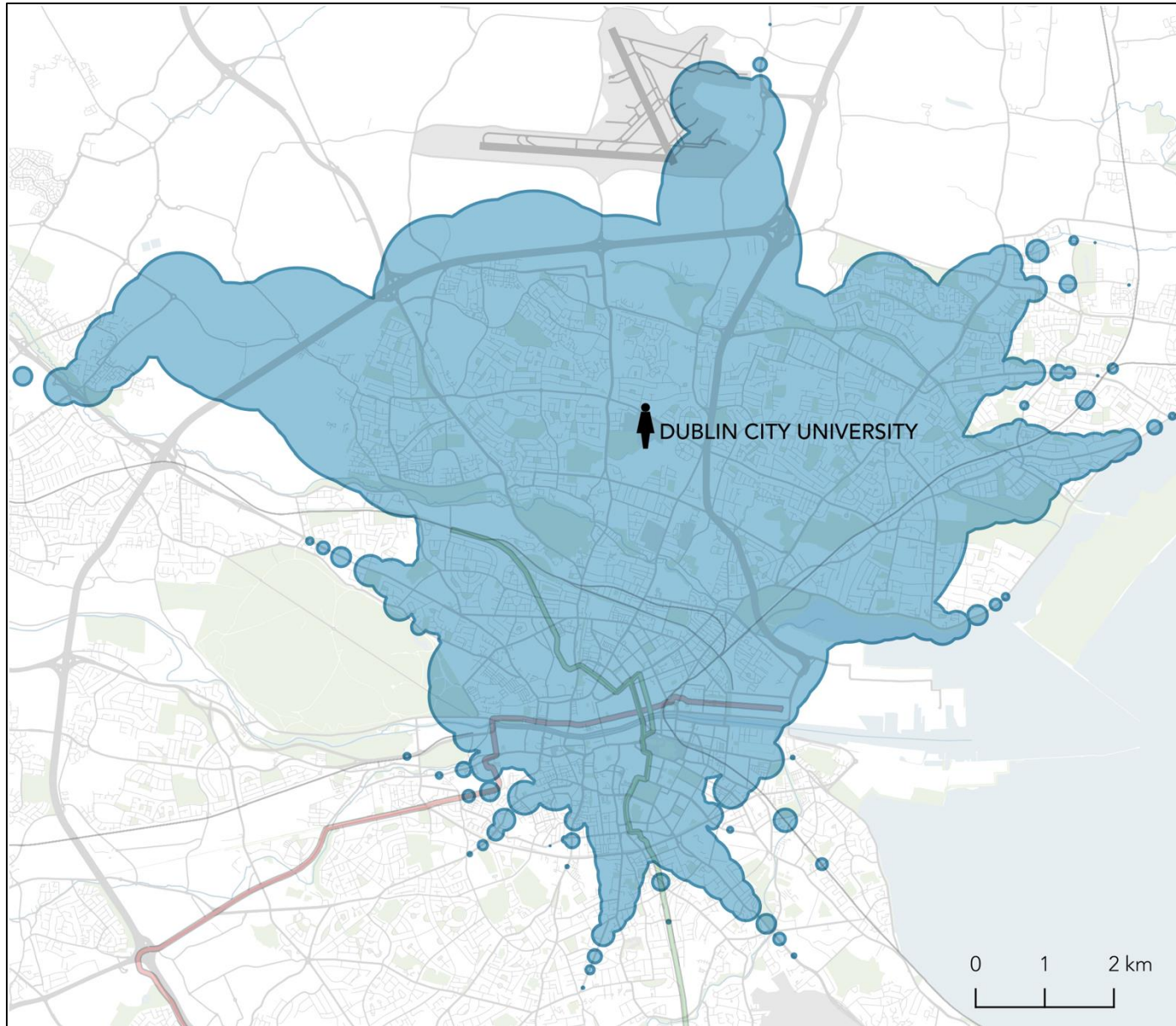
DUBLIN CITY UNIVERSITY

0 1 2 km

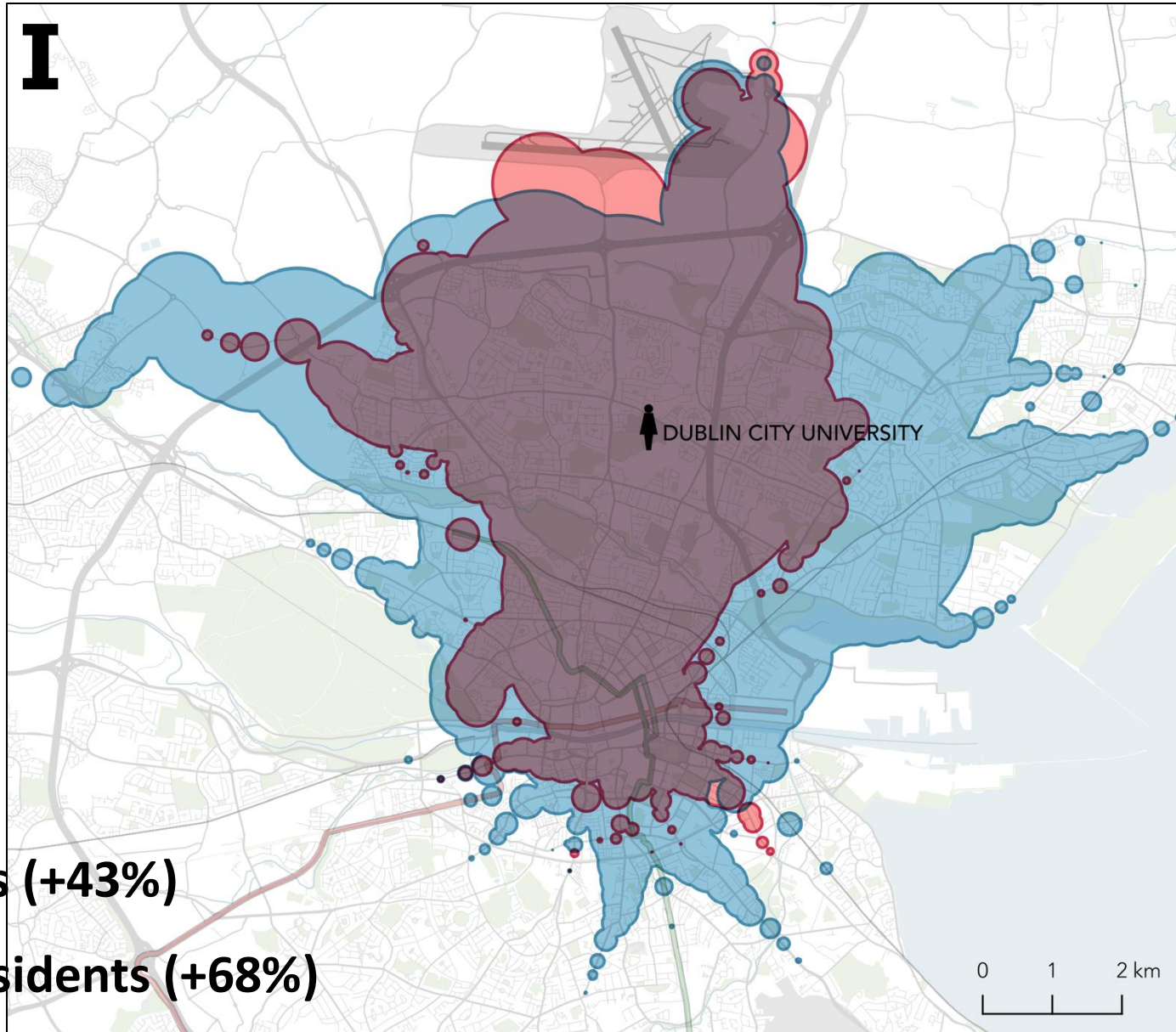
Existing Network



Proposed Network

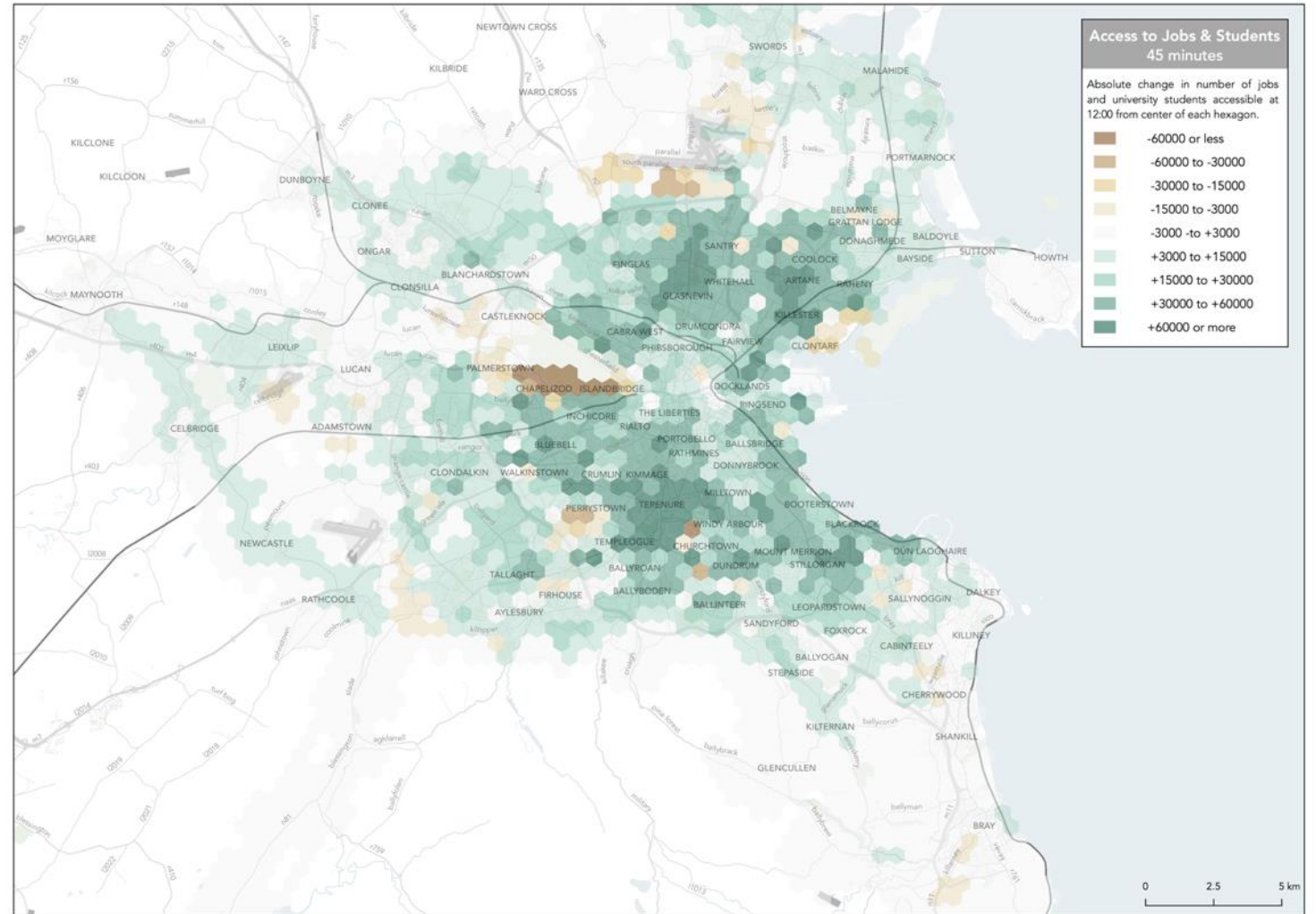


What can I reach?

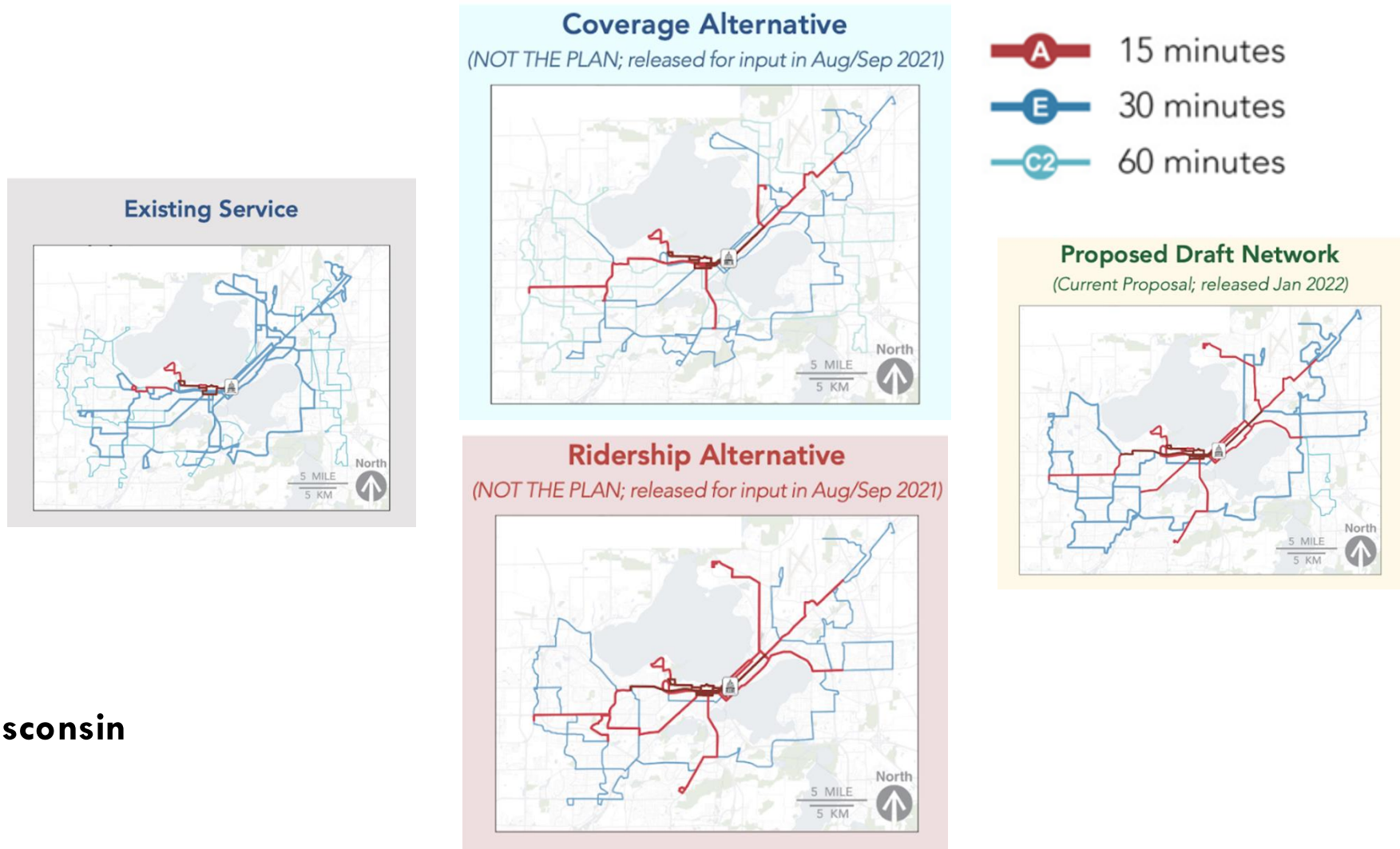


Citywide Access

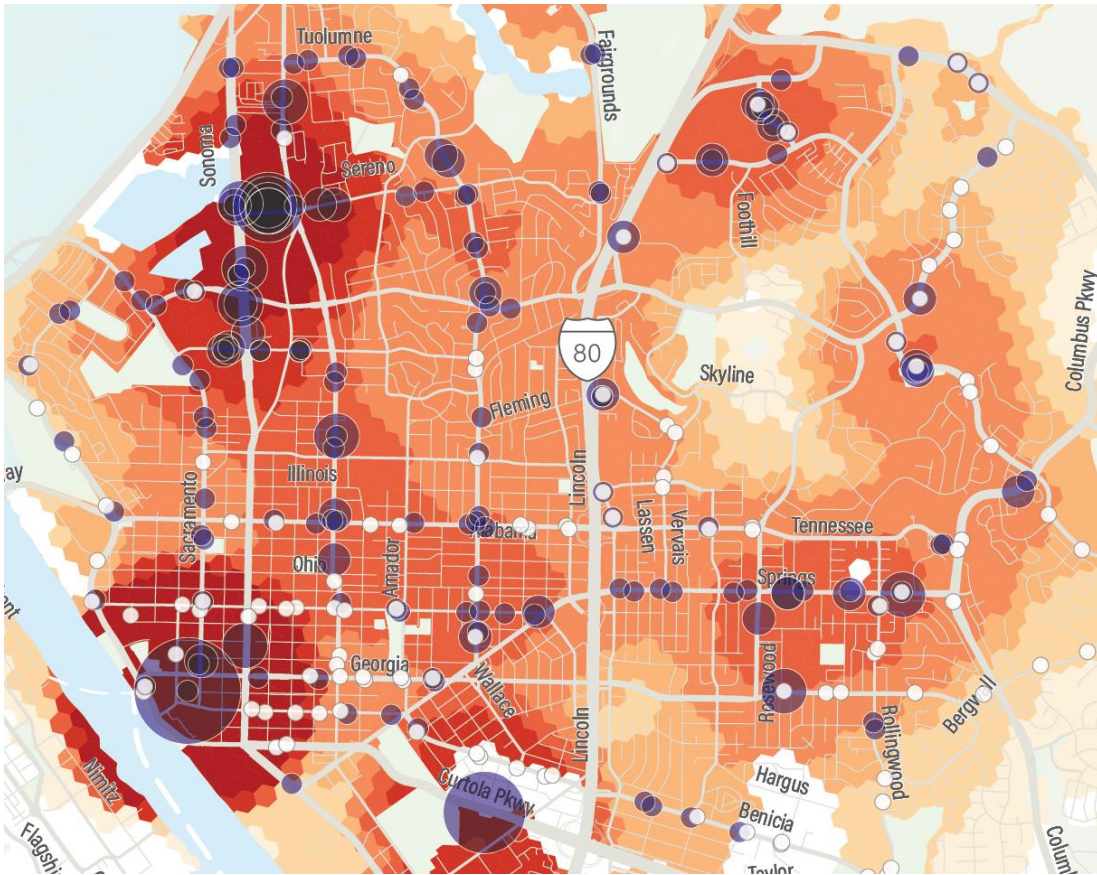
- On most of our projects, we make that same comparison sampling across an entire service area.
- Which areas benefit?
- Which don't?
- Whose access are we increasing, whose are we reducing?



These comparisons help us get to clear decisions.



**Madison, Wisconsin
Redesign
2022**



**Transit planning
today is inseparable
from data analysis.**

Individual experience is only a limited view of how the entire system works.

Every vehicle is a collection of sensors that produces a massive amount of data, constantly.

Open and closed data standards now exist for the most important things.

Every person that rides the system or contributes to its funding source is entitled to an opinion about it.

R is essential to our approach.

- We package and templatize as much as we can so that even the smallest agencies get the benefit of tools originally developed for big customers.
- Every analyst who joins our team learns to work in R at least to the level of running and debugging templated scripts.
- The R ecosystem has great tools for transit analysis, particularly the gtfstools and r5r packages developed by the team at the Institute for Applied Economic Research (IPEA) in Brazil.
- Over time, we've also developed a robust set of internal tools geared toward our specific approaches to common transit analysis problems.

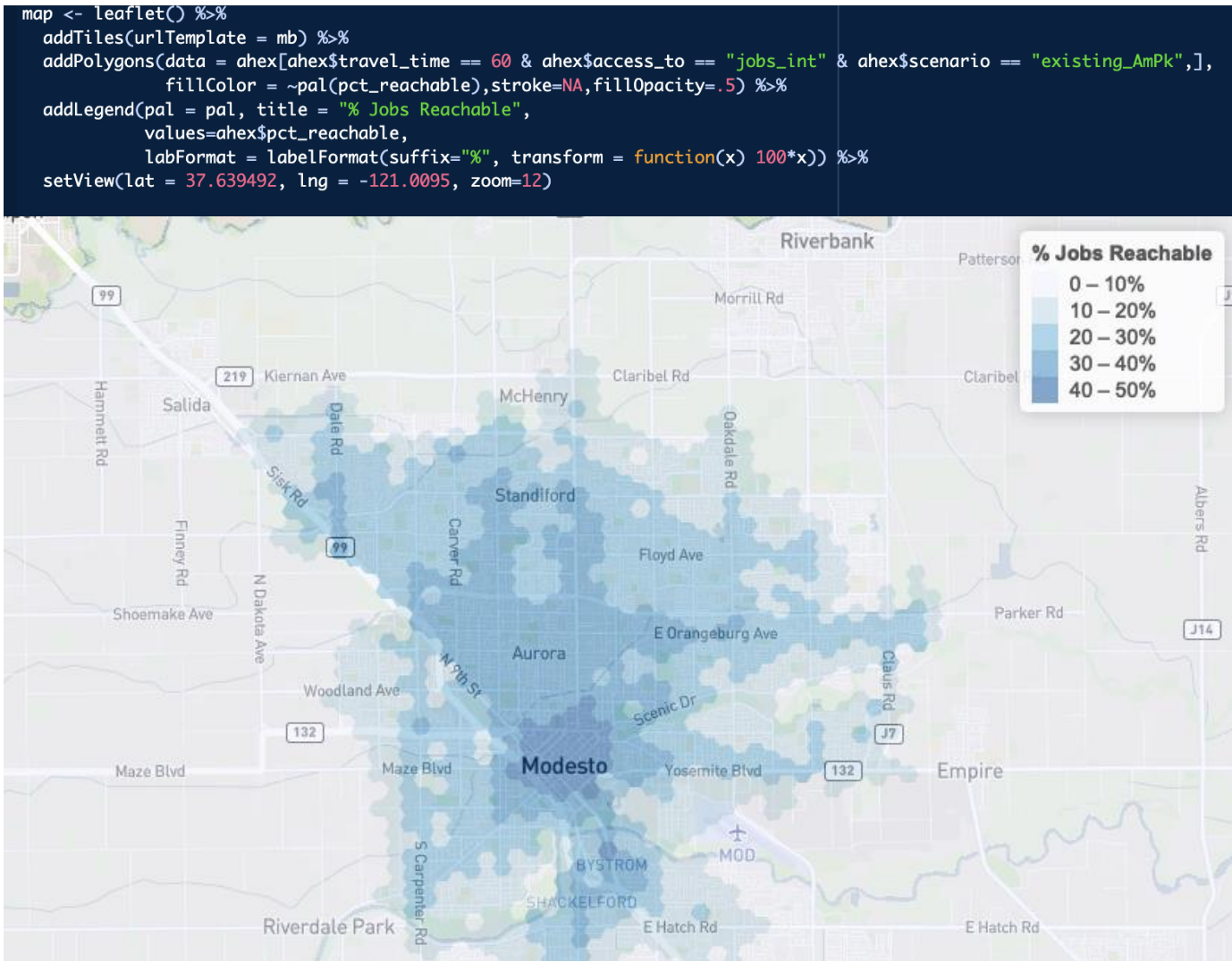
R Transit Analysis Toolbox

- Almost anything you want to do requires handling multiple large, messy datasets that feature both time and space dimensions.
- GTFS (General Transit Feed Specification) is the standard that makes almost everything else practical.
 - It's often the one thing that isn't messy!

trip_id	arrival_time	departure_time	stop_id	stop_sequence	stop_headsign	pickup_type
t200-b835-s12	05:12:00	05:12:00	1176	0	NA	0
t200-b835-s12	05:13:47	05:13:47	1177	107	NA	0
t200-b835-s12	05:14:50	05:14:50	1178	170	NA	0
t200-b835-s12	05:17:06	05:17:06	1179	306	NA	0
t200-b835-s12	05:19:00	05:19:00	1011	420	NA	0
t200-b835-s12	05:19:53	05:19:53	1180	473	NA	0
t200-b835-s12	05:32:00	05:32:00	1009	1200	NA	0
t200-b835-s12	05:36:00	05:36:00	1008	1440	NA	0
t200-b835-s12	05:54:04	05:54:04	1181	2524	NA	0
t200-b835-s12	05:55:39	05:55:39	1182	2619	NA	0
t200-b835-s12	05:57:00	05:57:00	429	2700	NA	0
t200-b835-s12	06:00:04	06:00:04	1184	2884	NA	0
t200-b835-s12	06:00:53	06:00:53	1185	2933	NA	0
t200-b835-s12	06:02:08	06:02:08	1012	3008	NA	0
t200-b835-s12	06:02:51	06:02:51	1186	3051	NA	0
t200-b835-s12	06:03:25	06:03:25	189	3085	NA	0
t200-b835-s12	06:04:10	06:04:10	1078	3130	NA	0
t200-b835-s12	06:12:00	06:12:00	1301	3600	NA	1
t276-b835-s12	06:30:00	06:30:00	1302	0	NA	0
t276-b835-s12	06:30:48	06:30:48	2	48	NA	0

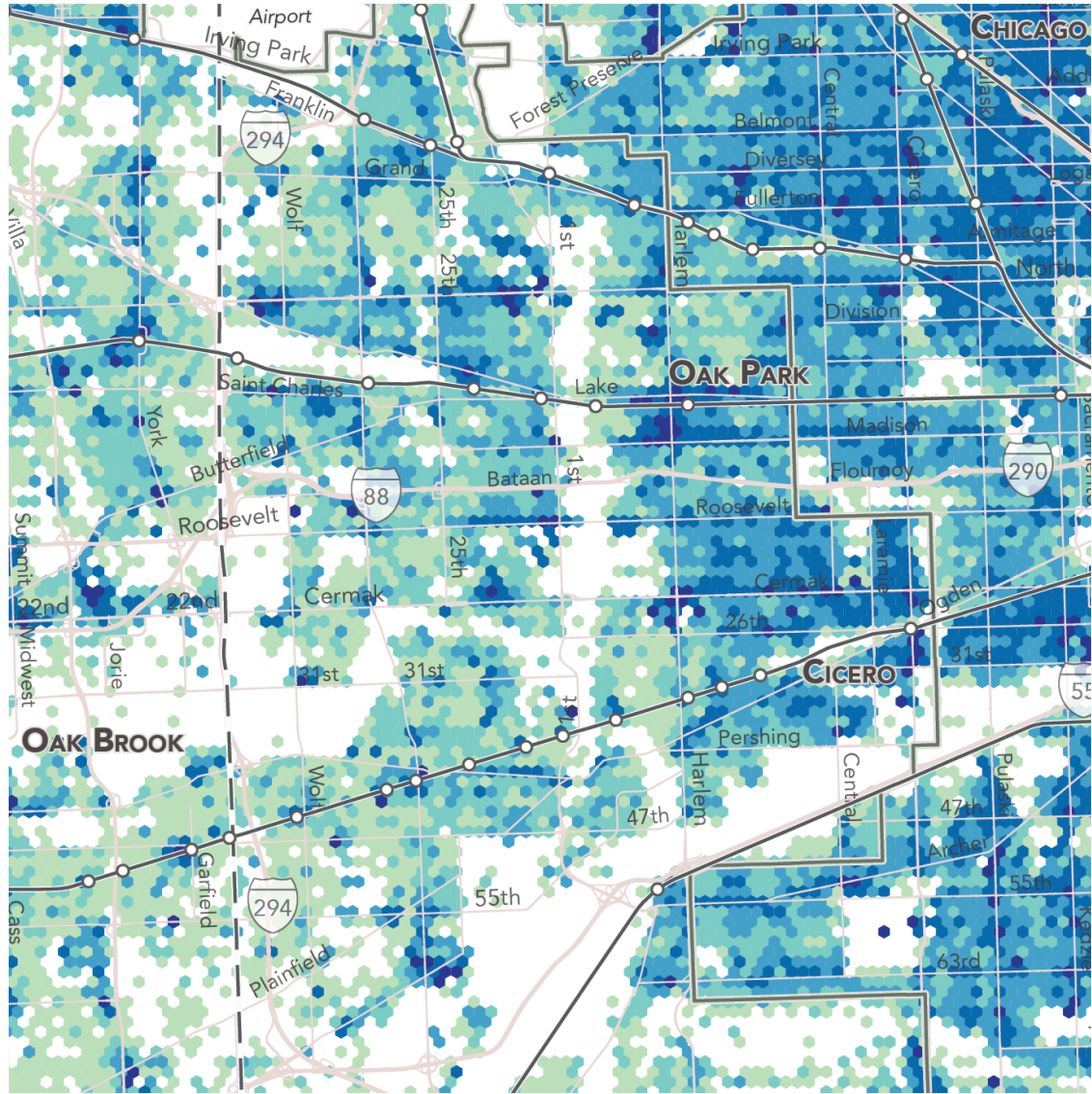
R Transit Analysis Toolbox

- Because transit is so spatial, efficient mapping is a crucial skill.
- Traditional GIS is slow; rarely practical for exploratory work, and a pain to iterate.
- A strong command of R mapping tools is helpful.

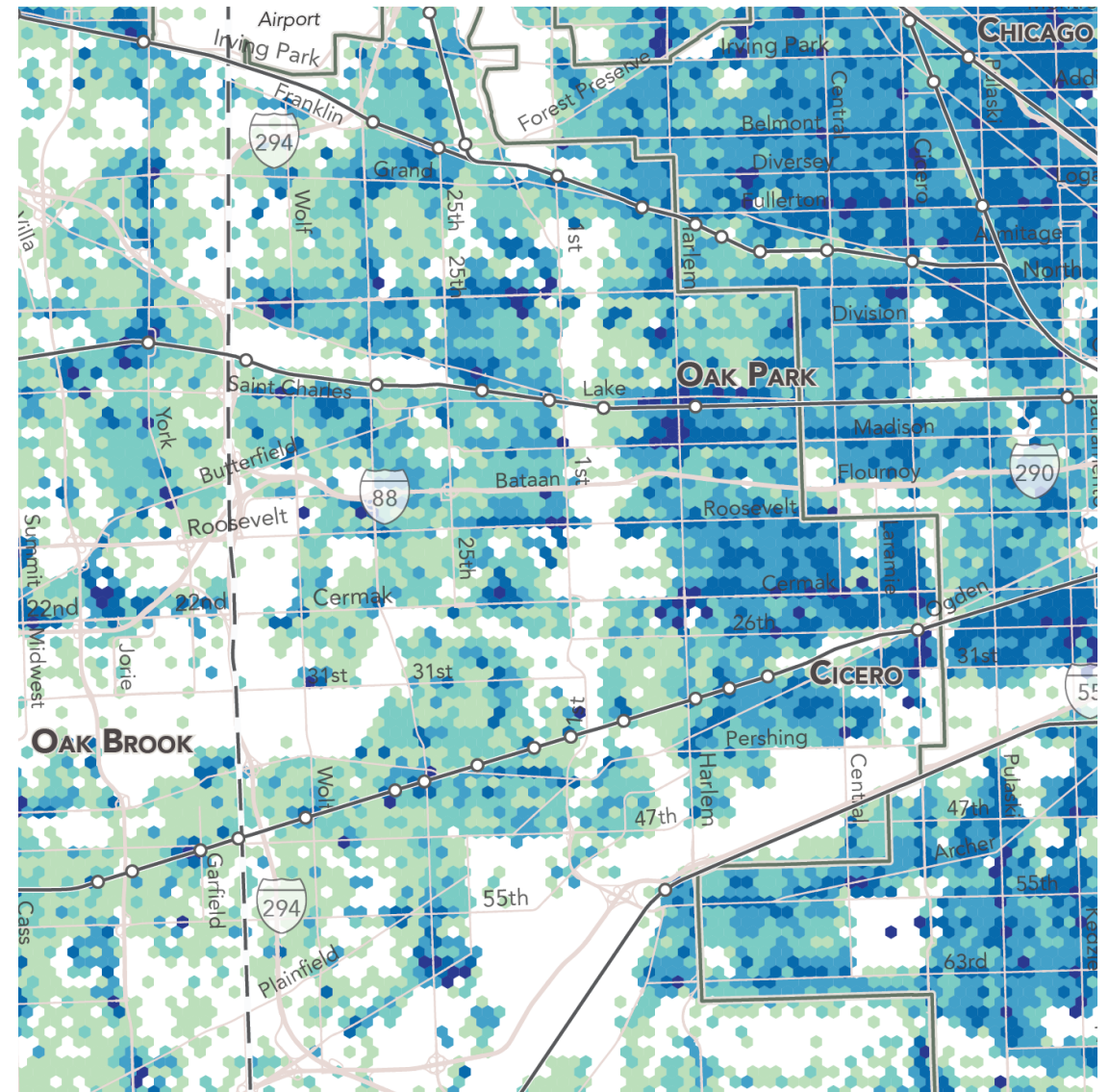


Travel Demand

- Census is essential– tidycensus, lehdr, ipumsr, etc.
- We're always working on our own tools for handling more granular proprietary mobility datasets.
- Since early on, data.table has been our best friend for these kinds of “fits on a laptop but just barely” applications.



- Census is essential— tidy`census`, `lehdr`, `ipumsr`, etc.
- We're always working on our own tools for handling more granular proprietary mobility datasets.
- Since early on, `data.table` has been our best friend for these kinds of “fits on a laptop but just barely” applications.



Development Approach

- **Experiment** – one-off analysis, responding to a specific client request.
- **Template** – the second time we do something, figure out how to do it efficiently and generically.
- **Package** – regularly review for processes for more robust development.

Thank you!

If you work on transit or are interested in learning more about what we do, don't hesitate to reach out!

Evan Landman
evan@jarrettwalker.com
Principal Associate
Jarrett Walker & Associates